

Although I think academia is a very noble endeavor, and I hope to pursue it, I am starting to feel like the "paper" format is becoming more of a burden to scientific advancement than an aid to disseminating ideas. If anything, I believe the future unit of academic discourse should be "the problem formulation", rather than a "problem + solution" -- let alone a full-fledged paper.

I am not the first to suggest a move away from the traditional paper. But I may be the first to propose so radical a notion as the mere "problem-formulation" as being sufficient for academic credit.

Why do I suggest this?

In my mind, the life of an applied mathematician for most of history has consisted of three parts: (1) the detection of some intriguing phenomenon in a real-world scientific workflow or industrial process;

(2) a distillation of said phenomenon into a toy mathematical model that captures its essential features (a necessarily lossy step); and

(3) a solution to the toy problem that (hopefully) provides guidance in the real-world scientific / industrial setting from whence it originated.

For much of my life, having grown up in various math circles, I identified as a person in camp 3, a "problem-solver". However, at a certain point in life, we realize that (at least for me...) there are many people (and nowadays, AI-chatbots...) more talented than us at the "problem-solving aspect". At such a stage, someone can either fall into despair, or they can choose to adjust their perspective.

That's why I suggest the "problem-formulation" as the crucial unit of scientific progress. Plus, it has the side-benefit of being more fun to engage in, and more inclusive of the broader community (if implemented in a web-based fashion, many many more people can be involved at the early stages of problem-formulation, whereas very few humans can solve an SDP [I certainly can't...]).

It's hard for me to defend this futuristic-vision against certain criticisms, such as accusations of intellectual-laziness (which may very well be true in my case...).

Nevertheless, I think what I'm saying is also true to some degree, as I am hearing a lot of complaints about the utter collapse of the quality of recent reviewing cycles.

I'll keep thinking about this. Perhaps we could even run a "mini-trial" of such a rapidly-iterative mode of scientific discovery.